

COST FUNCTIONS WITH AN AVOIDABLE FIXED COSTS

ECON101—INTRODUCTION TO MICROECONOMICS.

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Consider a firm with production function

$$q = f(L, K, Z),$$

where $L \geq 0$ is labor, $K \geq 0$ is capital, and

$$Z \in \{0, 1\}$$

is a discrete input. Interpret $Z = 1$ as leasing the facility required for production, and $Z = 0$ as not leasing it. Suppose

$$f(L, K, Z) = ZL^{1/2}K^{1/2}.$$

Notice that strictly positive levels of production, $q > 0$, are only possible when leasing the facility, i.e., $Z = 1$.

The price of labor is $w > 0$, the rental price of capital is $r > 0$, and the price of leasing the facility is $\ell_z > 0$.

Questions

- (a) Suppose the firm is in the short run. The lease has already been signed, so $Z = 1$, and capital is fixed at $\bar{K} > 0$. Derive the short-run cost function.
- (b) Suppose the firm is in the long run. The firm can choose L , K , and Z . Derive the long-run cost function.
- (c) Explain the sense in which the lease payment is a fixed cost in the long run, but not a sunk cost.

Solution

Since

$$f(L, K, Z) = ZL^{1/2}K^{1/2},$$

positive output is possible only if $Z = 1$.

Short-run cost

In the short run, $Z = 1$ and $K = \bar{K}$. Therefore,

$$q = L^{1/2}\bar{K}^{1/2}.$$

Solving for the amount of labor required to produce q ,

$$q^2 = L\bar{K},$$

so

$$L = \frac{q^2}{\bar{K}}.$$

Therefore, the short-run cost function is

$$C^{SR}(q; \bar{K}) = w \frac{q^2}{\bar{K}} + r \bar{K} + \ell_z.$$

In particular,

$$C^{SR}(0; \bar{K}) = r \bar{K} + \ell_z > 0.$$

Thus, in the short run, the firm pays the rental cost of fixed capital and the lease payment even if it produces zero output.

Long-run cost

In the long run, the firm chooses L , K , and Z . The long-run cost function is

$$C^{LR}(q) = \min_{L, K \geq 0, Z \in \{0, 1\}} wL + rK + \ell_z Z \quad \text{s.t.} \quad ZL^{1/2}K^{1/2} \geq q.$$

If $q = 0$, the firm can choose $L = K = 0$ and $Z = 0$. Hence

$$C^{LR}(0) = 0.$$

If $q > 0$, then the firm must choose $Z = 1$. The problem becomes

$$C^{LR}(q) = \ell_z + \min_{L, K \geq 0} wL + rK \quad \text{s.t.} \quad L^{1/2}K^{1/2} \geq q.$$

The constraint is equivalent to

$$LK \geq q^2.$$

At the cost-minimizing bundle, the constraint binds:

$$LK = q^2.$$

Using $K = q^2/L$, minimize

$$wL + r \frac{q^2}{L}.$$

The first-order condition is

$$w - r \frac{q^2}{L^2} = 0.$$

Therefore,

$$L^*(q) = q \sqrt{\frac{r}{w}},$$

and

$$K^*(q) = q \sqrt{\frac{w}{r}}.$$

Substituting into total variable cost,

$$wL^*(q) + rK^*(q) = wq \sqrt{\frac{r}{w}} + rq \sqrt{\frac{w}{r}} = 2q\sqrt{wr}.$$

Therefore,

$$C^{LR}(q) = \begin{cases} 0, & q = 0, \\ \ell_z + 2q\sqrt{wr}, & q > 0. \end{cases}$$

Interpretation

The lease payment ℓ_z is fixed because, conditional on producing any positive amount of output, it does not vary with q . For every $q > 0$, the firm must pay ℓ_z .

However, in the long run, the lease payment is avoidable. If the firm chooses not to produce, it can choose $Z = 0$ and pay no lease cost. Thus,

$$C^{LR}(0) = 0.$$

By contrast, in the short run, $Z = 1$ is already fixed, so the firm must pay ℓ_z even when $q = 0$. Hence,

$$C^{SR}(0; \bar{K}) = r\bar{K} + \ell_z > 0.$$

Therefore, fixed costs can exist in the long run, but long-run fixed costs are avoidable rather than sunk.